

PRAVIN KUMAR

Indian Institute of Information Technology Allahabad, U.P., India

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ACADEMIC PROFILE

Assistant Professor and Ph.D. scholar in medical image analysis, deep learning, and computer vision, with teaching experience across algorithms, data structures, Python programming, pattern recognition, and deep learning. Research interests include diabetic retinopathy screening, lesion segmentation, fundus image analysis, interpretable AI, transformers, generative AI, and reliable AI for healthcare.

ACADEMIC AND PROFESSIONAL EXPERIENCE

Sri Balaji University, Pune

Assistant Professor

Pune, Maharashtra

Nov 2024 – Present

- Teach **Design and Analysis of Algorithms** and **Pattern Recognition and Deep Learning**; prepare course plans, learning material, assignments, examinations, and evaluation rubrics.
- Serve as **University SPOC for Smart India Hackathon (SIH)**, coordinating institutional participation and communication with the Ministry of Education.
- Lead the **Techno Tuner Club** for 2025–2026, mentoring student technical initiatives, innovation activities, and applied computing projects.

Galgotias University

Assistant Professor

Greater Noida, U.P.

Aug 2023 – Oct 2024

- Delivered lectures and lab sessions in **Data Structures, Algorithms, and Python Programming**.
- Mentored students on coursework, programming practice, and academic project development.

Ispeck Digital Solutions Private Limited

Data Scientist

Kolkata, West Bengal

Mar 2023 – Jun 2023

- Built computer vision solutions for face recognition and YOLO-based personal protective equipment compliance monitoring.
- Supported applied deployment of deep learning models for site security, surveillance, and management workflows.

EDUCATION

Ph.D. in Information Technology, Medical Image Analysis

Indian Institute of Information Technology Allahabad, U.P., India

Pursuing

M.Tech in Human-Computer Interaction, Information Technology

Indian Institute of Information Technology Allahabad, U.P., India

Jul 2016 – Jul 2018

CGPA: 8.00

MCA B-Level, Information Technology

National Institute of Electronics and Information Technology, New Delhi, India

Aug 2012 – Jun 2015

DOEACC A-Level, Information Technology

National Institute of Electronics and Information Technology, New Delhi, India

Jan 2010 – Jun 2011

DOEACC O-Level, Information Technology

National Institute of Electronics and Information Technology, New Delhi, India

Aug 2008 – Dec 2009

Higher Secondary School, Bihar State Board

RLSA Mahavidyalaya Vaishali, Bihar, India

Mar 2007

School, Bihar State Board

RVMS Maker Saran, Bihar, India

Mar 2005

PUBLICATIONS AND MANUSCRIPTS

1. **Kumar, P.** and Singh, S.K., “Cross-Dataset Diabetic Retinopathy Grading with a Lesion-Anchored Multi-View Transformer and Ordinal-Distance Knowledge Distillation.” *ICT Express*.
2. **Kumar, P.** and Singh, S.K., “MSDAG-Net: A Multi-Scale Dual-Attention-Guided Network for Microaneurysm Segmentation in Fundus Images.” *ICT Express*.
3. **Kumar, P.** and Singh, S.K., “Lesion-Graph Ordinal Reasoning with Calibrated Deferral for Cross-Dataset Diabetic Retinopathy Triage.” *CVIP 2026*.

4. **Kumar, P.** and Singh, S.K., “LCK-CBM: Typed Lesion Keypoints for Interpretable Cross-Dataset Diabetic Retinopathy Grading.” *CVIP 2016*.

TEACHING, MENTORING, AND RESEARCH

- **Teaching Assistant, IIIT Allahabad** (*Aug 2019 – Mar 2023*): Deep Learning, Computer Graphics, Image and Video Processing, Data Compression, and C Programming.
- **Research Mentor, CVBL, IIIT Allahabad** (*Aug 2019 – Mar 2023*): Mentored interns and M.Tech students in computer vision and medical image analysis projects.
- Developed research workflows using Python, PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, and image-analysis pipelines.

LEADERSHIP AND SERVICE

- **Head**, Techno Tuner Club, Sri Balaji University, Pune *2025–2026*
- **University SPOC**, Smart India Hackathon, Sri Balaji University, Pune.
- **Treasurer**, IEEE UP Section, IIIT Allahabad *2019–2022*
- **Presidential Council Member and Ph.D. Representative**, Gymkhana, IIIT Allahabad *2020–2022*
- Assisted in organizing IEEE Signal Processing Society activities and ADASIVA summer schools *2018–2022*
- Assisted in organizing the 5th International Conference on Computer Vision and Image Processing (CVIP), IIIT Allahabad *2021*

TECHNICAL SKILLS

Programming	Python, C++, C
AI/ML	PyTorch, TensorFlow, scikit-learn, deep learning, computer vision, generative AI
Computer Vision	OpenCV, fundus image analysis, lesion segmentation, object detection, face recognition
Core CS	Data structures, algorithms, OOP, DBMS, pattern recognition
Tools	Git, NumPy, Pandas, LaTeX

ACHIEVEMENTS AND OUTREACH

- Secured All India Rank 4619 in GATE 2016 among more than 100,000 candidates.
- Attended the workshop on Advances in Deep Architectures for Signal, Image and Vision Applications (ADASIVA 2018).
- Lead digital education outreach through [CSwithPravin](#) *Mar 2020 – Present*